Aleksandr Kovalenko

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EDUCATION

ETH Zurich (Swiss Federal Institute of Technology) | *M.Sc. Computer Science, Systems & Networking* 2015–2017

- Graduated with distinction (GPA 3.9/4.0); thesis on distributed edge computing architectures achieved 47 citations in peer-reviewed venues
- President of Systems & Optimization Club (2016–2017), grew membership from 12 to 180+ students; organized 8 industry seminars with speakers from Google, Cloudflare, and Cisco

University of Amsterdam | *B.Sc. Computer Science with minor in Electrical Engineering* 2011–2015

- Summa cum laude (GPA 3.85/4.0); awarded University Excellence Scholarship for top 2% of cohort
- Led peer mentorship program for 35 first-year students; coordinated 4 hackathons with 200+ participants each, sponsoring €15K in prizes

EXPERIENCE

Cloudflare | *Staff Engineer, Edge Computing & Networking* 2021–Present

- Architected and shipped Cloudflare Workers at the Edge v3 runtime, reducing cold-start latency by 87ms (from 152ms → 65ms) and increasing throughput by 340% (from 8.2K to 35.8K req/s per edge location); impacts 1.2M+ monthly active deployments across 275+ data centers
- Led 12-person distributed systems team building next-generation edge caching layer; reduced cache miss ratio by 23% (from 18.4% → 14.2%) through predictive pre-warming algorithms, saving customers $8.3M annually in bandwidth costs
- Designed real-time telemetry pipeline collecting 2.1B events/day from edge network with sub-millisecond latency guarantees; achieved 99.98% uptime SLA and reduced debugging time for incident resolution by 64% (from 45min → 16min median)

Shopify | *Senior Engineer, Infrastructure & Edge Networks* 2019–2021

- Rebuilt checkout payment processing infrastructure for edge execution, reducing P99 latency by 340ms (from 650ms → 310ms) and increasing transaction throughput by 2.8x (from 18K → 50.4K TPS); unblocked Q4 2020 peak season handling 228% YoY traffic growth
- Owned geo-distributed consensus protocol for inventory synchronization across 80 edge locations with sub-50ms consistency guarantees; improved data freshness by 91% and reduced split-brain incidents from 12/month → 0 over 8-month period
- Mentored 5 junior engineers; 3 were promoted to senior within 18 months; conducted 40+ code reviews maintaining 99.7% test coverage across 450K lines of critical infrastructure code

Datadog | *Senior Software Engineer, Edge Agents & Observability* 2017–2019

- Engineered lightweight edge agents (8.2 MB footprint) deployed to 18,000+ customer environments; optimized CPU overhead by 62% through lock-free data structures and async I/O, reducing customer compute costs by $4.2M annually
- Designed distributed tracing system for edge-to-cloud observability capturing 340M traces/day with 99.9% delivery guarantees and variable 2-5s latency; shipped live tracing feature reducing MTTR by 31% (from 23min → 16min average)
- Built cost-analysis module preventing 67% of user-initiated scaling errors through predictive modeling; saved customers $12.7M in wasted infrastructure spend over 18-month deployment

Wise (formerly TransferWise) | *Software Engineer, Backend & Payment Systems* 2015–2017

- Implemented high-performance currency routing engine processing 340K transactions/day with 99.97% uptime; reduced transaction settlement time by 4.2 hours through optimized batch processing, unblocking £2.8B additional monthly volume
- Owned microservices migration from monolith affecting 12 dependent teams; coordinated rollout of 26 services maintaining zero downtime; achieved 8.4x improvement in P99 latency (from 340ms →

40ms) for payment APIs
- Improved fraud detection ML model accuracy from 89.2% → 94.7% through feature engineering; reduced false positives by 58%, preventing $340K in legitimate transaction blocks

PROJECTS

EdgeMesh: Distributed Control Plane Framework | *Go, Rust, Protobuf, gRPC, eBPF, Kubernetes* 2019–Present
- Open-source edge networking framework with 3,200+ GitHub stars and 120K+ monthly downloads; enables sub-millisecond inter-node communication across heterogeneous edge clusters; benchmarks show 2.1x throughput improvement over Istio (47K vs 22K req/s) with 71% lower memory footprint
- Implemented novel eBPF-based packet steering achieving 94% reduction in context-switching overhead; 340+ contributing members from Cloudflare, Azure, and AWS contributing optimizations; adopted by 18 production platforms processing 8.2B events/day

Quantum: Adaptive Edge Caching System | *Python, C++, Redis, TensorFlow, CUDA* 2021–Present
- ML-driven predictive caching system reducing cache misses by 34% through lightweight neural networks (8.2KB model size); 1,400+ GitHub stars with adoption across 6 major CDNs; inference latency of 1.2ms at p95 ensures real-time cache decisions
- Achieved 340% ROI in customer deployments by reducing bandwidth costs $2.1M/year; open-sourced with comprehensive benchmarking suite; won 'Best Infrastructure Tool' at Open Source Infrastructure Summit 2022

RapidRoute: Global Load Balancing Simulator | *Rust, WebAssembly, TypeScript, WebGL, Kubernetes* 2020–2021
- Interactive browser-based simulator for geo-distributed load balancing algorithms; 890 GitHub stars; enables testing of 40+ algorithms across 250+ synthetic topologies with microsecond-precision timing; 15K monthly active users including engineers from Netflix, Stripe, and Uber
- Validated theoretical edge routing algorithms achieving 23% latency reduction vs production systems; benchmarked on realistic traces of 2.1B requests/day; published findings in 2 peer-reviewed systems conferences

EXTRACURRICULAR

USENIX ATC & ACM SIGCOMM | *Speaker & Session Chair (2019–2023)* 2019–2023
- Delivered 5 talks on edge computing architectures and distributed systems reaching 2,400+ live attendees and 80K+ YouTube views; 'Scaling Edge Networks to 275+ Points of Presence' talk (2022) cited in 34+ subsequent research papers
- Chaired 3 conference sessions covering networking and infrastructure; peer-reviewed 28 papers for USENIX ATC, ACM SIGCOMM, and NSDI spanning edge computing, distributed consensus, and observability

Open Source Mentorship Program (Linux Foundation & CNCF) | *Maintainer & Senior Mentor* 2018–Present
- Maintained 6 critical open-source projects with 7,200+ combined GitHub stars; mentored 23 early-career engineers with 18 (78%) advancing to senior roles within 3 years; reviewed 2,400+ pull requests with average turnaround of 8.2 hours
- Established code quality standards and testing infrastructure increasing CI/CD coverage from 62% → 98%; fostered inclusive community attracting 340+ international contributors across 42 countries; organized annual summit with 180+ attendees

TECHNICAL SKILLS

Languages – Go, Rust, Python, C++, TypeScript, Java, Bash, Protobuf

Technologies – Kubernetes, gRPC, eBPF, TensorFlow, Redis, PostgreSQL, Cassandra, Prometheus, Jaeger, WebAssembly, CUDA, Docker, Envoy, QUIC, HTTP/3

Tools – Git, Terraform, Datadog, PagerDuty, Grafana, ELK Stack, AWS (EC2, Lambda, CloudFront), GCP (Compute Engine, Cloud CDN), GitHub Actions, Bazel, perf, flamegraph, strace